

Exercise. 4.8

$$\phi(t, \vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}} e^{-ip \cdot x} + b_{\vec{p}}^\dagger e^{ip \cdot x})$$

$$\phi^\dagger(t, \vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (b_{\vec{p}} e^{-ip \cdot x} + a_{\vec{p}}^\dagger e^{ip \cdot x})$$

Under the parity transformation we get $P \phi(t, \vec{x}) P^\dagger = \phi(t, -\vec{x})$

We also note that under parity transformation

$$\vec{x} \rightarrow -\vec{x} \quad \text{and} \quad \vec{p} \rightarrow -\vec{p}$$

$$P \phi(t, \vec{x}) P^\dagger = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} P (a_{\vec{p}} e^{-ip \cdot x} + b_{\vec{p}}^\dagger e^{ip \cdot x}) P^\dagger = \phi(t, -\vec{x})$$

$p \cdot x$ is invariant under parity transformation since $\vec{p} \cdot \vec{x} \rightarrow (-\vec{p}) \cdot (-\vec{x}) = \vec{p} \cdot \vec{x}$

$$P \phi(t, \vec{x}) P^\dagger = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (P a_{\vec{p}} P^\dagger e^{-ip \cdot x} + P b_{\vec{p}}^\dagger P^\dagger e^{ip \cdot x}) \quad \parallel \quad P a_{\vec{p}} e^{-ip \cdot x} P^\dagger = P a_{\vec{p}} P^\dagger P e^{-ip \cdot x} P^\dagger$$

Now we relate this to $\phi(t, -\vec{x})$

$$\phi(t, -\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}} e^{-i(Et + \vec{p} \cdot \vec{x})} + b_{\vec{p}}^\dagger e^{i(Et + \vec{p} \cdot \vec{x})})$$

$$\phi(t, -\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{-\vec{p}} e^{-ip \cdot x} + b_{-\vec{p}}^\dagger e^{ip \cdot x})$$

We can write this in simpler way by changing $\vec{p} \rightarrow -\vec{p}$

From the condition of $P \phi(t, \vec{x}) P^\dagger = \phi(t, -\vec{x})$ we get

$$P a_{\vec{p}} P^\dagger = a_{-\vec{p}} \quad \text{and} \quad P b_{\vec{p}}^\dagger P^\dagger = b_{-\vec{p}}^\dagger$$

By taking adjoint of these we get:

$$(P a_{\vec{p}} P^\dagger)^\dagger = P a_{\vec{p}}^\dagger P^\dagger = (a_{-\vec{p}})^\dagger = a_{-\vec{p}}^\dagger$$

Similarly

Similar

$$(P b_{\vec{p}}^\dagger P^\dagger)^\dagger = P b_{\vec{p}} P^\dagger = (b_{-\vec{p}})^\dagger = b_{-\vec{p}}^\dagger$$

So, under the parity transformation we get

$$P a_{\vec{p}} P^\dagger = a_{-\vec{p}} \quad \& \quad P a_{\vec{p}}^\dagger P^\dagger = a_{-\vec{p}}^\dagger$$

$$P b_{\vec{p}} P^\dagger = b_{-\vec{p}} \quad \& \quad P b_{\vec{p}}^\dagger P^\dagger = b_{-\vec{p}}^\dagger$$

Now we look into the time reversal

$$T \phi(t, \vec{x}) T^\dagger = \phi(-t, \vec{x})$$

$$\rightarrow T \phi(t, \vec{x}) T^\dagger = T \left[\int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}} e^{i p \cdot x} + b_{\vec{p}}^\dagger e^{i p \cdot x}) \right] T^\dagger$$

From Peskin and Schroeder (right after eq. (3.133)) we get $T e^{+iHt} = e^{-iHt} T$

which means $T e^{+iEt} = e^{-iEt} T$. Similarly we can extend this

to $T e^{-i\vec{p} \cdot \vec{x}} = e^{+i\vec{p} \cdot \vec{x}} T$ since like t , $\vec{p}$ changes sign under time reversal

As such we get $T e^{-i p \cdot x} T^\dagger = e^{i p \cdot x}$. Using this we get

$$T \phi(t, \vec{x}) T^\dagger = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (T a_{\vec{p}} T^\dagger e^{i p \cdot x} + T b_{\vec{p}}^\dagger T^\dagger e^{-i p \cdot x})$$

Meanwhile:

$$\phi(-t, \vec{x}) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}} e^{-i(-Et - \vec{p} \cdot \vec{x})} + b_{\vec{p}}^\dagger e^{i(-Et - \vec{p} \cdot \vec{x})})$$

$$\phi(-t, \vec{x}) \Rightarrow \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}} e^{i(Et + \vec{p} \cdot \vec{x})} + b_{\vec{p}} e^{-i(Et + \vec{p} \cdot \vec{x})}) = \int \frac{d^3 p}{(2\pi)^3} (a_{-\vec{p}} e^{i p \cdot x} + b_{-\vec{p}}^\dagger e^{-i p \cdot x})$$

Where we have again done transformation $\vec{p} \rightarrow -\vec{p}$. Since $T \phi T^\dagger = \phi(-t, \vec{x})$

we get

$$T a_{\vec{p}} T^\dagger = a_{-\vec{p}}^\dagger \quad \& \quad T b_{\vec{p}}^\dagger T^\dagger = b_{-\vec{p}}^\dagger \quad \text{now we take adjoint to get}$$

$$(T a_{\vec{p}} T^\dagger)^\dagger = T a_{\vec{p}}^\dagger T^\dagger = a_{-\vec{p}}^\dagger \quad \& \quad \text{similarly} \quad T b_{\vec{p}} T^\dagger = b_{-\vec{p}}$$

So under the time reversal we get:

$$T a_{\vec{p}} T^\dagger = a_{-\vec{p}}^\dagger \quad \& \quad T a_{\vec{p}}^\dagger T^\dagger = a_{-\vec{p}}$$

$$T b_{\vec{p}} T^\dagger = b_{-\vec{p}} \quad \& \quad T b_{\vec{p}}^\dagger T^\dagger = b_{-\vec{p}}^\dagger$$

Now we do charge conjugation

$$C \phi(t, x) C^\dagger = \phi^*(t, x) = \phi^\dagger(t, x) \quad \text{since } \phi \text{ is a complex scalar field}$$

$$C \phi(t, \vec{x}) C^\dagger = C \left[\int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}} e^{-ip \cdot x} + b_{\vec{p}}^\dagger e^{ip \cdot x}) \right] C^\dagger = \phi^\dagger(t, \vec{x})$$

Under charge conjugation p or x don't change so

$$C \phi(t, \vec{x}) C^\dagger = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (C a_{\vec{p}} C^\dagger e^{-ip \cdot x} + C b_{\vec{p}}^\dagger C^\dagger e^{ip \cdot x}) = \phi^\dagger(t, \vec{x})$$

Now we write $\phi^\dagger(t, \vec{x})$

$$\phi^\dagger(t, \vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E}} (a_{\vec{p}}^\dagger e^{ip \cdot x} + b_{\vec{p}} e^{-ip \cdot x})$$

We match the correct exponents

$$\Rightarrow C a_{\vec{p}} C^\dagger = b_{\vec{p}} \quad \text{and} \quad C b_{\vec{p}}^\dagger C^\dagger = a_{\vec{p}}^\dagger$$

By taking a adjoint we get $C a_{\vec{p}}^\dagger C^\dagger = b_{\vec{p}}^\dagger$ & $C b_{\vec{p}} C^\dagger = a_{\vec{p}}$

So, under the charge conjugation we get

$$C a_{\vec{p}} C^\dagger = b_{\vec{p}} \quad \& \quad C a_{\vec{p}}^\dagger C^\dagger = b_{\vec{p}}^\dagger$$

$$C b_{\vec{p}} C^\dagger = a_{\vec{p}} \quad \& \quad C b_{\vec{p}}^\dagger C^\dagger = a_{\vec{p}}^\dagger$$

Now we need to find the transformation properties of the current

$$J^\mu = i(\phi^* \partial^\mu \phi - \partial^\mu \phi^* \phi) \quad \text{under } C, P \text{ \& } T$$

$$P J^\mu P^\dagger = i(P \phi^* P^\dagger P \partial^\mu P^\dagger P \phi P^\dagger - P \partial^\mu P^\dagger P \phi^* P^\dagger P \phi P^\dagger) \quad \parallel \quad P^\dagger P = 1 \text{ was used}$$

Under parity transformation we get $P \partial^\mu P^\dagger = (-1)^\mu \partial^\mu$

$$\begin{aligned} \Rightarrow P J^\mu P^\dagger &= i(-1)^\mu (P \phi^* P^\dagger \partial^\mu P \phi P^\dagger - \partial^\mu P \phi^* P^\dagger P \phi P^\dagger) \\ &= i(-1)^\mu (\phi^*(t, -\vec{x}) \partial^\mu \phi(t, \vec{x}) - \partial^\mu \phi^*(t, -\vec{x}) \phi(t, \vec{x})) \end{aligned}$$

$$P J^\mu P^\dagger = (-1)^\mu i(\phi^* \partial^\mu \phi - \partial^\mu \phi^* \phi)(t, -\vec{x})$$

// here we took arguments of ϕ^* & ϕ out for less clutteriness.

Under time reversal we get $T \partial^\mu T^\dagger = -(-1)^\mu \partial^\mu$

$$\begin{aligned} \Rightarrow T J^\mu T^\dagger &= -i(T \phi^* T^\dagger \underbrace{T \partial^\mu T^\dagger}_{= (-1)^\mu \partial^\mu} T \phi T^\dagger - T \partial^\mu T^\dagger T \phi^* T^\dagger T \phi T^\dagger) \\ &= +i(-1)^\mu (\phi^* \partial^\mu \phi - \partial^\mu \phi^* \phi)(-t, \vec{x}) \end{aligned}$$

// T acts on c-number by conjugating it:
 $T c = c^* T$

under charge conjugation we have $C\phi C^\dagger = \phi^*$, $C\phi^* C^\dagger = \phi$ and $C\partial^\mu C^\dagger = \partial^\mu$

$$\begin{aligned} \Rightarrow C J^\mu C^\dagger &= C (i(\phi^* \partial^\mu \phi - \partial^\mu \phi^* \phi)) C^\dagger \\ &= i (C\phi^* C^\dagger C\partial^\mu C^\dagger C\phi C^\dagger - C\partial^\mu C^\dagger C\phi^* C^\dagger C\phi C^\dagger) \\ &= i (\phi \partial^\mu \phi^* - \partial^\mu \phi \phi^*) \end{aligned}$$

Exercise 4.8,

We assume that the Dirac spinor has an internal $SU(3)$ invariance,

$$\Psi = \begin{pmatrix} q_r \\ q_b \\ q_g \end{pmatrix}, \quad \Psi \rightarrow \Psi' = e^{i\theta^a t^a} \Psi = U \Psi, \quad \text{where}$$

$t^a/2$ are the eight generators of $SU(3)$ Lie algebra.

If all θ^a , $a = 1, \dots, 8$ are constant the free Lagrangian

$$\mathcal{L} = \bar{\Psi} (i \not{\partial} - m) \Psi \quad \text{is invariant}$$

Now we consider local transformation $U = U(x)$ i.e. $\theta^a = \theta^a(x)$ and replace ordinary derivative with a covariant derivative

$$\partial_\mu \rightarrow \partial_\mu + ig A_\mu \quad \text{where } \hat{A}_\mu = A_\mu^a t^a \text{ is a matrix.}$$

The Lagrangian is then $\mathcal{L} = \bar{\Psi} (i \not{D} - m) \Psi$, where $\not{D} = \gamma^\mu (\partial_\mu + ig A_\mu)$

We require this to be invariant under local $SU(3)$ transformation

$$\Rightarrow \mathcal{L} \rightarrow \mathcal{L}' = U(x) \mathcal{L} U^\dagger(x) = \mathcal{L}$$

$$\mathcal{L}' = U \bar{\Psi} (i \not{D} - m) \Psi U^\dagger \quad \parallel \quad U U^\dagger = 1 \quad \text{since } U \text{ is unitary}$$

$$\begin{aligned} &= U \bar{\Psi} U^\dagger (i U \not{D} U^\dagger - U m U^\dagger) U \Psi U^\dagger \\ &= \bar{\Psi} (i \gamma^\mu (U \partial_\mu U^\dagger + ig U A_\mu U^\dagger) - m) \Psi \end{aligned} \quad \left\{ \begin{array}{l} \Psi \text{ is invariant under } SU(3) \\ \Rightarrow U \Psi U^\dagger = \Psi \\ U \bar{\Psi} U^\dagger = \bar{\Psi} \end{array} \right.$$

$$\mathcal{L}' = \bar{\Psi} (i \gamma^\mu (\partial_\mu - i t^a (\partial_\mu \theta^a) + ig A'_\mu) - m) \Psi$$

$$\begin{aligned} \partial_\mu (U^\dagger \Psi) &= (\partial_\mu U^\dagger) \Psi + U^\dagger \partial_\mu \Psi \\ \partial_\mu e^{\overleftarrow{-i\theta^a t^a}} &= e^{-i\theta^a t^a} (-i t^a) (\partial_\mu \theta^a) \end{aligned}$$

For $\mathcal{L}'$ to be same as $\mathcal{L}$ we require

$$-i t^a (\partial_\mu \theta^a) + ig A'_\mu = ig A_\mu \quad \Rightarrow \quad A'_\mu = A_\mu + \frac{t^a}{g} (\partial_\mu \theta^a)$$

So the gauge field A_μ transforms as

$$\underline{A_\mu \rightarrow A_\mu + \frac{1}{g} \partial_\mu \theta(x)}$$

Exercise 4.9

We have the Lagrangian

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \Phi)^2 - \frac{1}{2} M^2 \Phi^2 + \frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2 - \mu \Phi \phi \phi$$

We assume $2m < M$. The decay rate of Φ is given by

$$\int d\Gamma = \frac{1}{2M} \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \frac{1}{4E_1 E_2} |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}(P_\Phi - p_1 - p_2)$$

We assume known $|\mathcal{M}|^2 = 4\mu^2$. Here p_1 is momentum of ϕ_1 -particle
 p_2 is momentum of ϕ_2 -particle.

$$\delta^{(4)}(P_\Phi - p_1 - p_2) = \delta(E_\Phi - E_1 - E_2) \delta^{(3)}(\vec{P}_\Phi - \vec{p}_1 - \vec{p}_2)$$

Let's use the centre of mass frame: $E_\Phi = M$ & $\vec{P}_\Phi = 0$

$$\Rightarrow \int d\Gamma = \frac{4\mu^2}{2M} \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^2} \frac{1}{4E_1 E_2} \delta^{(3)}(\vec{p}_1 - \vec{p}_2) \delta(M - E_1 - E_2)$$

Integrating over $d^3 p_2$ we get

$$\int d\Gamma = \frac{\mu^2}{2M} \int \frac{d^3 p_1}{(2\pi)^2} \frac{1}{E_1^2} \delta(M - 2E_1)$$

$$\int d\Gamma = \frac{\mu^2}{2M} \int \frac{d^3 p_1}{(2\pi)^2} \frac{1}{\vec{p}_1^2 + m^2} \delta(M - 2\sqrt{\vec{p}_1^2 + m^2})$$

The previous delta-function fixed $E_1 = E_2$ since they are energies of particles with same mass & momentum
 $E_1 = \vec{p}_1^2 + m^2$

We can write this delta-function as $\delta(M^2 - 4m^2 - 4\vec{p}_1^2)$
 $= \delta(\vec{p}_1^2 - (\frac{M^2}{4} - m^2))$

We evaluate this integral in spherical coordinates $d^3 p_1 = |p|^2 dp d\Omega$ // here $|p| = |\vec{p}|$

$$\int d\Gamma = \frac{\mu^2}{2M(2\pi)^2} \int_{4\pi} d\Omega \int dp \frac{|p|^2}{|p|^2 + m^2} \delta(|p|^2 - (\frac{M^2}{4} - m^2))$$

To evaluate the delta function we use the relations: $\delta(g(x)) = \frac{\delta(x-x_0)}{|g'(x_0)|}$, $g(x_0) = 0$

For our deltafunction we get

$$\delta(|p|^2 - (\frac{M^2}{4} - m^2)) = \frac{\delta(|p| - \sqrt{M^2/4 - m^2})}{2\sqrt{M^2/4 - m^2}}$$

$$\begin{aligned} g'(x) &= 2x \\ \text{for} \\ g(x) &= x^2 - (\frac{M^2}{4} - m^2) \\ x_0 &= \sqrt{\frac{M^2}{4} - m^2} \quad \text{positive} \\ \text{root since } x &\text{ is only positive} \\ &= |p| \end{aligned}$$

$$\Rightarrow \Gamma = \int d^3p = \frac{\mu^2}{2\pi M} \int d|p| \frac{|p|^2}{|p|^2 + m^2} \cdot \frac{\delta(|p| - \sqrt{M^2/4 - m^2})}{2\sqrt{M^2/4 - m^2}}$$

$$= \frac{\mu^2}{4\pi M} \frac{M^2/4 - m^2}{M^2/4} \cdot \frac{2}{M\sqrt{1 - \frac{4m^2}{M^2}}}$$

$$\Gamma = \frac{\mu^2}{2\pi M^2} \left(1 - \frac{4m^2}{M^2}\right) \left(\sqrt{1 - \frac{4m^2}{M^2}}\right)^{-1} = \frac{\mu^2}{2\pi M^2} \sqrt{1 - \frac{4m^2}{M^2}}$$

The lifetime of the particle is $\tau = \frac{1}{\Gamma}$

$$\Rightarrow \tau = \frac{2\pi M^2}{\mu^2} \left(\sqrt{1 - \frac{4m^2}{M^2}}\right)^{-1}$$